

Course: B.Tech

Branch : Civil Engineering

Semester : III

Subject Code & Name: BTCVC306_Y19 Engineering Geology

Max Marks: 60

Date:09/02/2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Mars

Q.1 Objective type questions. (Compulsory Question)

12

- 1 What is the scientific study of the Earth, its materials, and processes called ? 1
a. Astronomy b. Biology c. Geology d. Chemistry
- 2 Fossils are most commonly found in which type of rocks? 1
a. Igneous b. Metamorphic c. Sedimentary d. All types equally
- 3 The continuous process where rocks change from one type to another is called the: 1
a. Plate b. Rock Cycle c. Carbon Cycle d. Water Cycle
Tectonic Cycle
- 4 Which mineral group is abundantly found in the Earth's crust? 1
a. Carbonates b. Micas c. Feldspars d. Oxides
(Silicates)
- 5 Which process is NOT involved in the formation of sedimentary rocks? 1
a. Weathering b. Deposition c. Metamorphism d. Consolidation/Lithification
- 6 Which geological structure involves the bending or curving of rock layers? 1
a. Fault b. Joint c. Fold d. Unconformity
- 7 Which geological structure represents a discontinuity in the deposition of 1
sedimentary layers?
a. Fault b. Joint c. Fold d. Unconformity
- 8 What is the term for a geologic layer that stores and transmits significant amounts of 1
water?
a. Aquiclude b. Aquitard c. Aquifer d. Aquifuge
- 9 Reservoir silting is primarily caused by 1

	a. Evaporation rates	b. High flow rates	c. Erosion and transport of fine sediments from the catchment area	d. Algal blooms	
10	Which geological structure can cause tunnel failure ?				1
	a. Horizontal bedding planes	b. Cross-faults and folded structures causing unequal stress	c. Massive, unfractured rock	d. Soft clay seams	
11	What instrument is used to measure the angle of inclination of rock layers (dip)?				1
	a. Clinometer.	b. Compass.	c. Seismograph.	d. Gravimeter.	
12	What is the primary objective of preserving geological cores?				1
	a. To increase their monetary value	b. To ensure the accuracy of future core analysis by preventing alteration of properties	c. To make them lighter for transport	d. To make them look aesthetically pleasing	
Q.2 Solve the following.					12
	A) Write a note on interior of the earth with neat labeled diagram.				6
	B) Write note on transportation by wind.				6
Q.3 Solve the following.					12
	A) What is clastic and non- clastic sedimentary rocks and their structures?				6
	B) Describe in brief about importance of minerals and their use in modern industries.				6
Q.4 Solve Any Two of the following.					12
	A) Define fold and describe classification of folds with neat labeled diagrams.				6
	B) Define and describe joints and classification with its role in civil engineering.				6
	C) What are the factors generally considered by an engineer while deciding use of stone in construction jobs?				6

Q.5 Solve Any Two of the following.	12
A) What is ground water? Describe the various zones of ground water.	6
B) Give details of importance of geological structures of reservoir site.	6
C) Illustrate important geological requirements considered in the selection of a dam site.	6
 Q.6 Solve Any Two of the following.	 12
A) Describe procedure of measurement and calculation of Rock Quality Designation (RQD) and its uses in construction.	6
B) Describe various types of drilling techniques in detail.	6
C) Write a note on objectives of preliminary geological survey in civil engineering structures.	6

***** End *****